YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

CANDIDATE NAME			
CG		INDEX NO	

BIOLOGY 9744/01

Paper 1 Multiple Choice **17 Sep 2021**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid/tape.
Write your name, index no. and CG on this cover page and Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

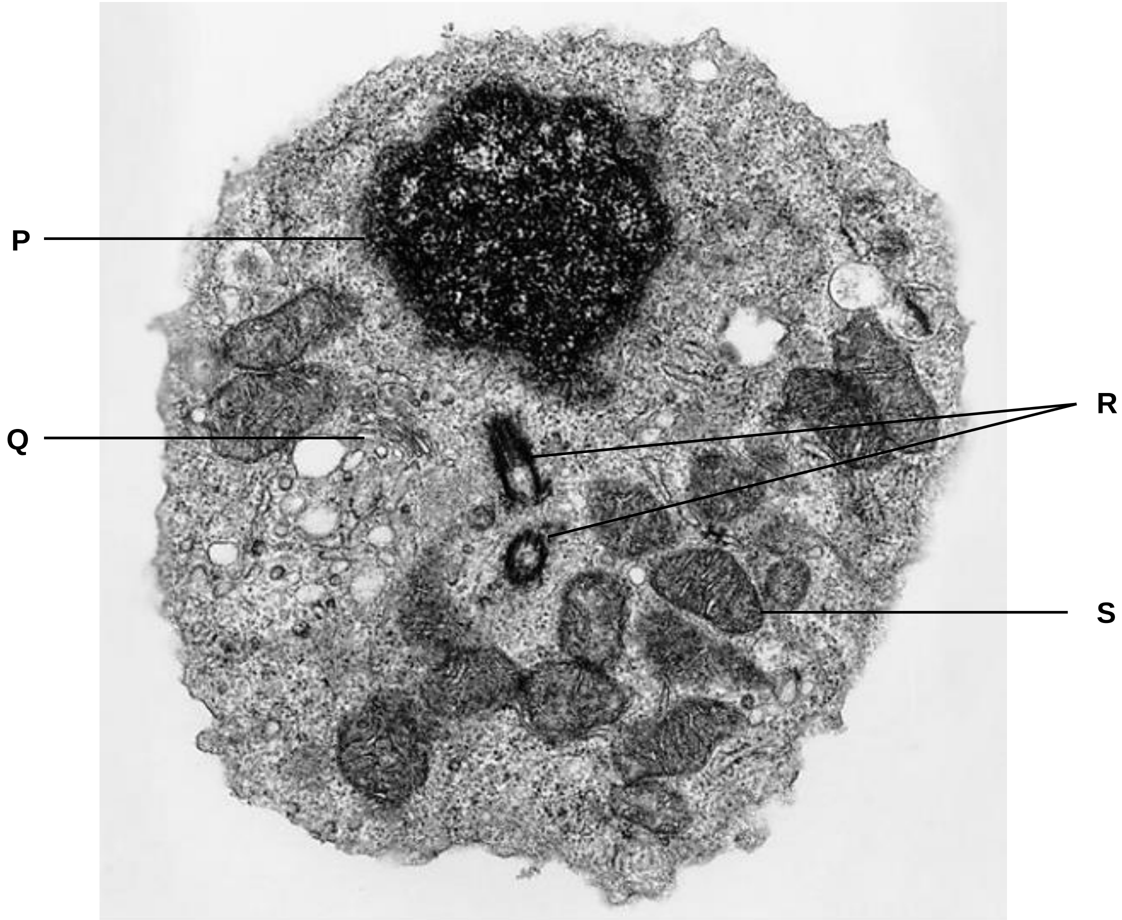
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **21** printed pages and **1** blank page.

1 The diagram shows an electron micrograph of a cell.



Which row correctly identifies the roles of the structures **P** to **S**?

	P	Q	R	S
A	DNA synthesis	polypeptide synthesis	spindle fibre formation	absorption of light
B	polypeptide synthesis	protein sorting	contains hydrolytic enzymes	triose phosphate synthesis
C	mRNA synthesis	protein packaging	DNA replication	glucose synthesis
D	contains genetic material	protein modification	cell division	ATP synthesis

2 Two of the isomers of glucose, α-glucose and β-glucose, can be condensed in living cells to form a variety of compounds with properties suited to different functions.

Which role correctly describes a polymer of glucose found in some living organisms?

	Molecular structure	isomer	bonds	function
A	branched chains	β-glucose	1, 4	energy reserve
B	coiled chains	α-glucose	1,4	structural support
C	branched chains	α-glucose	1,4 and 1,6	energy reserve
D	straight chains	β-glucose	1,6	structural support

3 The statements are about the loading of haemoglobin with oxygen.

1 As the concentration of oxygen increases, the loading of oxygen increases.

2 Haemoglobin that has not bound to oxygen has a higher affinity for 2,3-bisphosphoglycerate (2,3-BPG) than for oxygen. 2,3-BPG is present in all red blood cells.

3 Binding of some oxygen to haemoglobin increase the affinity of haemoglobin for additional oxygen.

4 Binding of oxygen to haemoglobin causes any bound carbon dioxide or bound 2,3-BPG to be expelled.

5 Low pH reduces the affinity of haemoglobin for oxygen.

What explains the effect of a low concentration of oxygen on the loading of oxygen by haemoglobin?

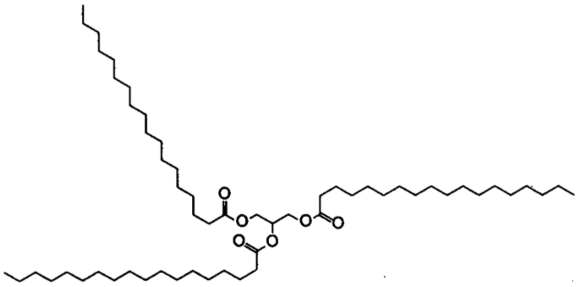
A The affinity of haemoglobin for oxygen decreases at low concentrations of oxygen since carbon dioxide replaces oxygen, causing the pH to fall.

B Carbon dioxide remains bound to amino acids in the haemoglobin molecule and therefore makes these sites unavailable for the binding of oxygen.

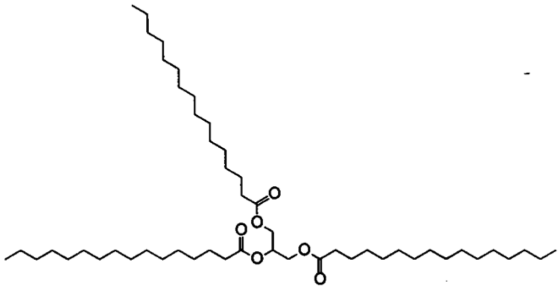
C At low oxygen concentrations, oxygen is unable to compete with 2,3-BPG for binding sites with haemoglobin.

D At low oxygen concentrations, carbon dioxide expels 2,3-BPG from haemoglobin, decreasing the affinity of haemoglobin for oxygen.

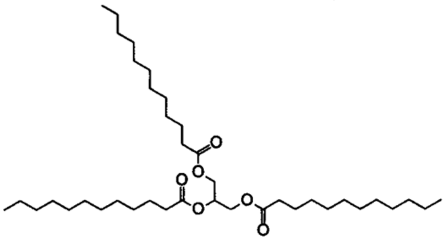
4 The formulae and melting points of five triglycerides are shown in the diagram. Each triglyceride contains three identical fatty acids.



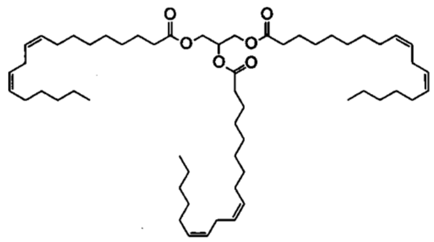
tristearin 72 °C



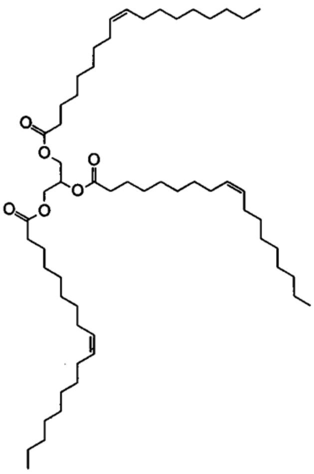
tripalmitin 65.5 °C



trilaurin 46 °C



trilinolein –13 °C



triolein –4 °C

Which two structural features of the molecules make the melting point higher?

	number of double bonds	length of fatty acid chains
A	fewer	longer
B	fewer	shorter
C	more	longer
D	more	shorter

5 Cells are the basic unit of living organisms.

What are the advantages to a multicellular organism of being made of many cells?

1 Disabling one cell is unlikely to affect the functioning of the whole organism.

2 Cells of different types are genetically different, decreasing the risk of expressing harmful recessive alleles.

3 Cells are able to specialise for specific functions, increasing efficiency.

4 Damaged cells of one type can be replaced easily by differentiation of cells of any other type.

A 1, 2 and 3

B 1, 3 and 4

C 1 and 3 only

D 2 and 4 only

6 The descriptions below are of nucleic acids in eukaryotes.

1 A polynucleotide of variable length formed by base pairing.

2 A small polynucleotide with a specific three-dimensional shape.

3 A large polynucleotide with a specific shape associated with proteins.

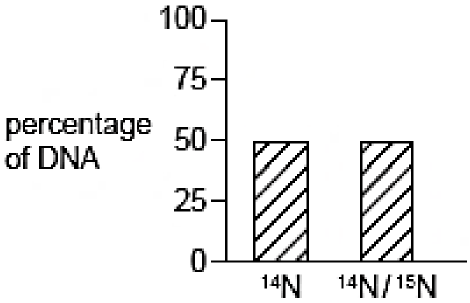
4 A large polynucleotide with super coiled sections associated with proteins.

Which row correctly matches each description to its function?

	Stores coded information	Carries coded information	Carries specific amino acids	Provides a site for protein synthesis
A	3	4	1	2
B	3	4	2	1
C	4	1	2	3
D	4	3	1	2

7 Bacteria were grown in a medium containing ^{15}N. After several generations, all of the DNA contained ^{15}N. Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N. The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N.

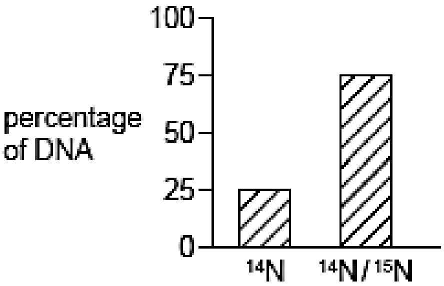
Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N. The bacteria were allowed to divide twice. The graph shows the percentages of ^{14}N and ^{15}N in the DNA of these bacteria.



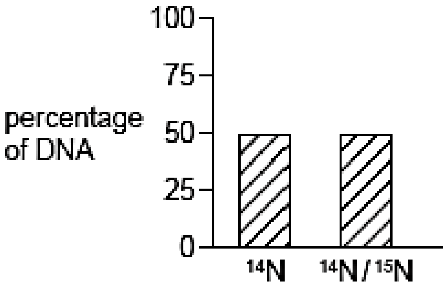
Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N. The bacteria were allowed to divide three times.

What would be the percentages of ^{14}N and ^{15}N in the DNA extracted from these bacteria?

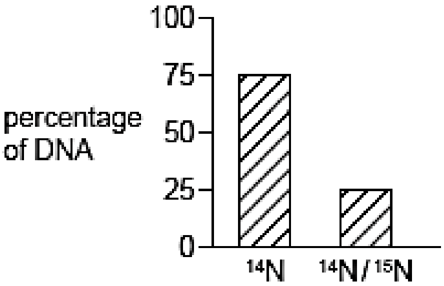
A



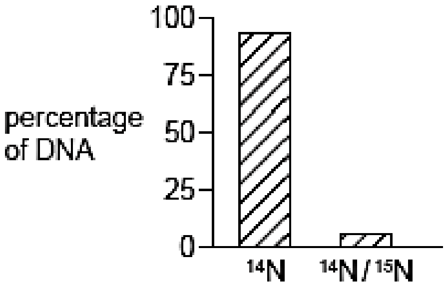
B



C



D



8 Scientists have modified the cellular machinery of *Escherichia coli* to allow synthesis of proteins from mRNA codons consisting of four nucleotides, rather than three. By significantly increasing the number of different codons, this provides the potential to synthesise novel proteins that incorporate synthetic amino acids.

For example, the four nucleotide codon AGGA was assigned to code for the synthetic amino acid p-azido-L-phenylalanine.

Which parts of the cellular machinery of *E. coli* need to be modified before synthesis of these novel proteins can occur?

A mRNA and RNA polymerase

B ribosomes and tRNA

C RNA polymerase and ribosomes

D tRNA and nucleotides

9 Messenger RNA is isolated by passing homogenized tissue through a fractionating column. The column consists of short lengths of uracil nucleotides attached to a solid support medium.

The mRNA that passes through the column cannot be translated as they break up into fragments. The mRNAs that attach to the column can be separated by application of sodium hydroxide.

Which statement is true of the mRNA in this experiment?

1 Complementary base-pairing occurs between adenine and uracil bases in the column.

2 The mRNA that pass through the fractionating column probably lack poly A tails.

3 The mRNA that attach to the fractionating column can be translated.

4 The mRNA that attach to the fractionating column probably lack poly A tails.

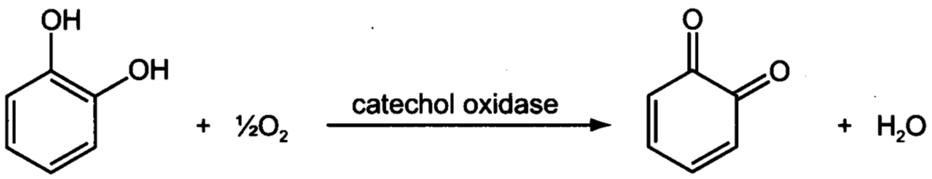
A 1 and 4 only

B 1, 2 and 3

C 1, 2 and 4

D All of the above

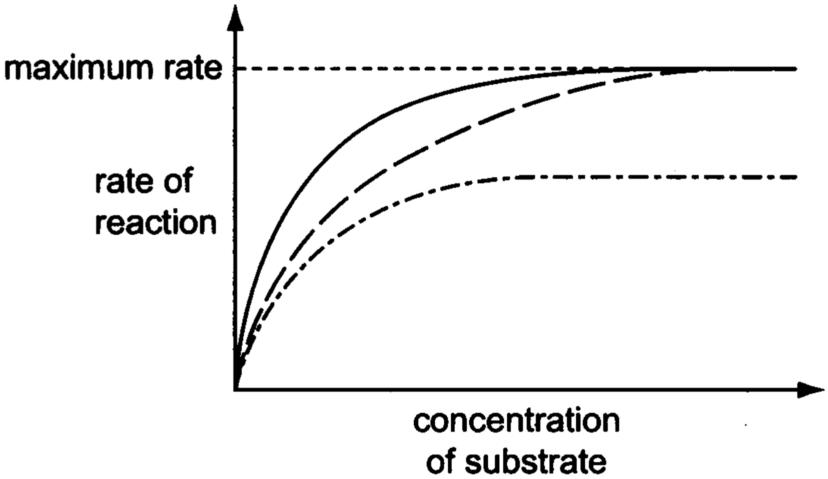
10 Catechol is oxidised to a quinone, as shown in the equation, resulting in darkening of peeled fruits.



Catechol oxidase is an enzyme which is inhibited by parahydroxybenzoic acid (PHBA) which is structurally similar to catechol.

It is also inhibited by phenylthiourea (PTU) which binds to a copper atom in the enzyme. The copper atom is essential for the oxidative activity.

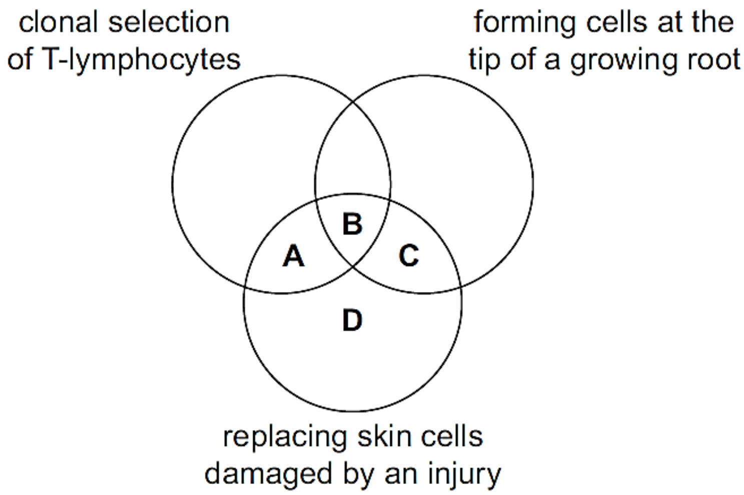
The graph shows the rate of the reaction with and without inhibitors.



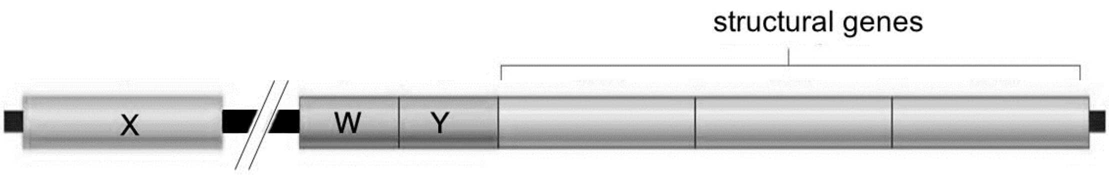
Which is the correct key to the curves?

	With PHBA	With PTU	uninhibited
A	– · – · – · – · –	– – – – – – – –	————————
B	————————	– · – · – · – · –	– – – – – – – –
C	– – – – – – – –	– · – · – · – · –	————————
D	– – – – – – – –	————————	– · – · – · – · –

11 Which processes involve mitosis?



12 The diagram shows a length of DNA responsible for metabolising lactose in prokaryotes. A mutation occurred in **X** such that its protein product became non-functional.

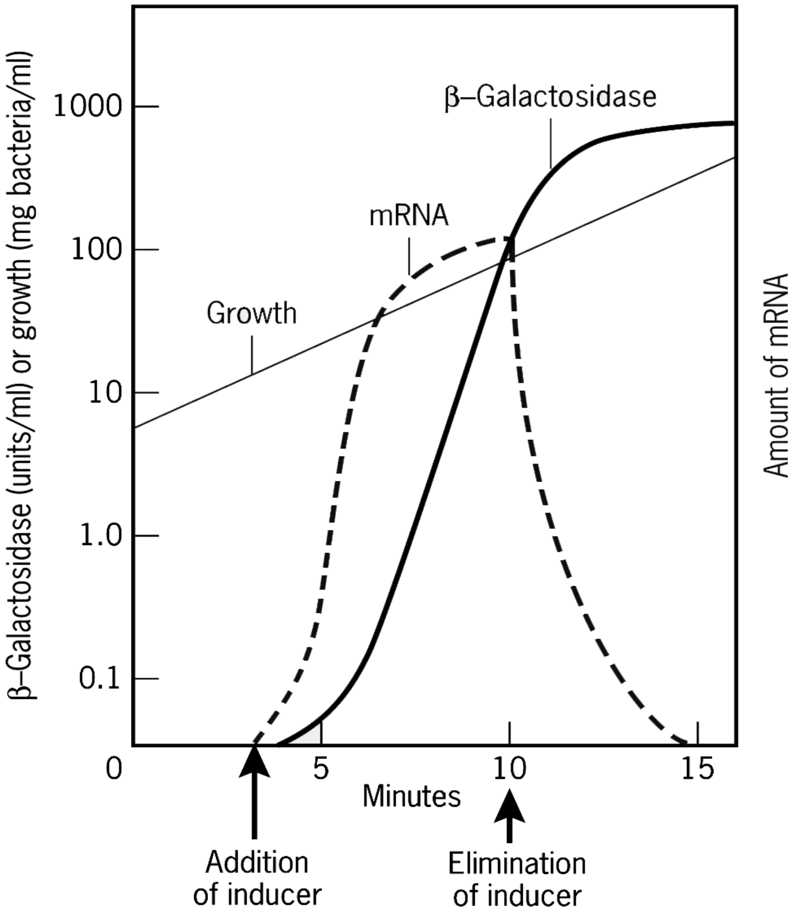


What is a possible outcome of the mutation if the bacteria cell is grown in a culture medium containing only lactose?

	Sequence bound by functional protein **X**	Positive control of gene regulation	Negative control of gene regulation	Transcription of structural genes
A	W	absent	present	no
B	Y	absent	present	no
C	W	present	absent	yes
D	Y	present	absent	yes

13 Prokaryotes such as *Escherichia coli* can be grown using culture medium containing essential elements and nutrients.

The regulation of gene expression in prokaryotes can be examined by studying the kinetics of the inducible *lac* operon. The figure below shows the changes in the levels of β-galactosidase mRNA and enzyme, as well as the growth of bacteria when lactose (inducer) was added to and then removed from the bacterial culture.



Which of the following conclusions can be inferred from the figure?

A A rapid decrease in the level of β-galactosidase mRNA upon removal of lactose suggests that RNA polymerase is still able to bind to the promoter and transcribe the gene coding for β-galactosidase at a much lower rate.

B The amount of β-galactosidase enzyme levels off upon removal of lactose, suggesting that the enzyme is stable and not degraded immediately even though it is no longer needed.

C The β-galactosidase mRNA is polycistronic as it contains coding sequences for structural genes such as lactose permease and lactose transacetylase.

D The brief delay in the increase of β-galactosidase mRNA and enzyme levels upon addition of lactose is due to the need for conversion of lactose to allolactose, which will then bind to the *lac* repressor.

14 The dengue virus is spherical, membrane-bound with a similar reproduction cycle as the influenza virus. However, unlike the influenza virus, its genetic material consists of one single positive sense strand of RNA.

Using the information above, which one of the following statements about the reproduction cycle of dengue virus is **false**?

A RNA-dependent RNA polymerase is required to complete its cycle.

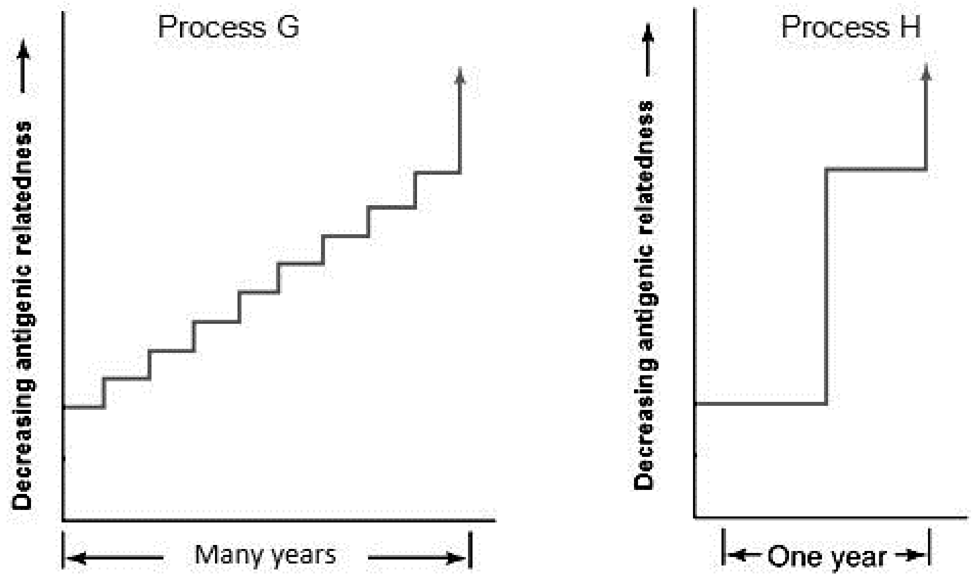
B The viral envelope fuses with the cell surface membrane of the host cell.

C The viral genome is directly used for translation of viral protein.

D Uncoating process involves fusion with endosome membrane.

15 Influenza infections are difficult to eliminate because of the antigenic changes that occur in haemagglutinin (HA), a surface glycoprotein of influenza virus. These changes are responsible for the formation of influenza viral strains with different levels of virulence.

The figures below depict two processes (G and H) that contribute to the decreasing antigenic relatedness of HA between different viral strains over time.



Which conclusions about processes G and H are valid?

1 G occurs due to a single point mutation in the gene coding for HA protein.

2 H may occur due to the genetic re-assortment of RNA genomes of human and avian influenza viruses.

3 G and H could only occur in an individual host who is simultaneously infected with at least two different strains of influenza virus.

4 The immune system of a host functions as a selective force for new antigenic variants of HA formed via process G.

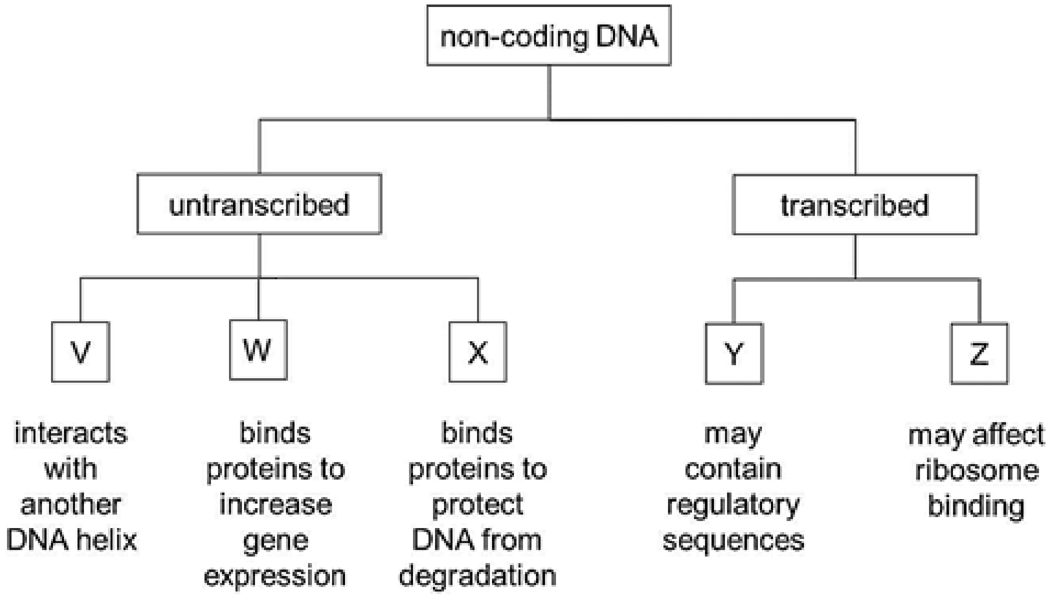
A 1 and 2

B 1 and 3

C 2 and 3

D 2 and 4

16 The diagram shows the classification of several portions of eukaryotic DNA that do not encode RNA or protein.



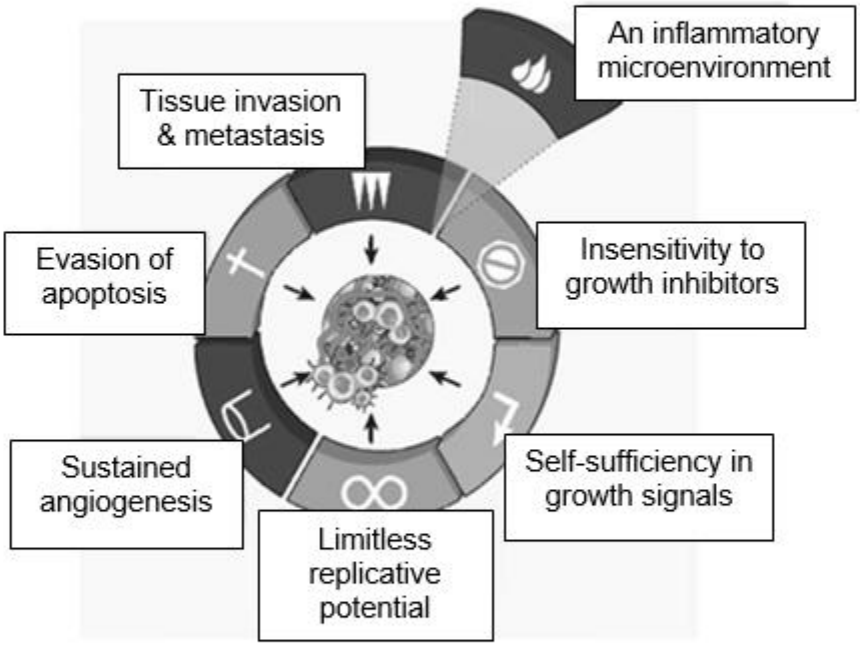
Which combination correctly identifies the type of eukaryotic DNA?

	V	W	X	Y	Z
A	centromere	intron	telomere	enhancer	5’ UTR
B	centromere	enhancer	telomere	intron	5’ UTR
C	intron	enhancer	telomere	5’ UTR	centromere
D	telomere	enhancer	intron	5’ UTR	centromere

17 If there are three control elements (**1**, **2** and **3**) regulating gene expression, which of the following shows a probable relationship for tissue specific gene expression within pancreatic cells (denoted **P**) and eye cells (denoted **E**) in eukaryotes?

	Contain gene for insulin	Contain genes for crystalline (lens protein)	Gene expression regulated by		
			Control element 1	Control element 2	Control element 3
A	**P** & **E**	**P** & **E**	**P** & **E**	**P** only	**E** only
B	**P** & **E**	**P** & **E**	**P** & **E**	**P** & **E**	**P** & **E**
C	**P** only	**E** only	**P** only	**P** & **E**	**P** only
D	**P** only	**E** only	**E** only	**P** only	**E** only

18 The figure below illustrates the hallmarks of cancer.



Which of the following statements correctly describe the changes in cancer cells?

1 Limitless potential of cancer cells to replicate often results in the accumulation of chromosomal mutations in many cancer cells.

2 Cancer cells can overproduce signal molecules so that they become self-sufficient in growth signals.

3 Angiogenesis is the result of expression of oncogenes in a cell line that produces blood vessels.

4 Loss-of-function mutation in tumour suppressor genes contribute to tissue invasion and metastasis.

A 1, 2 and 4

B 2, 3 and 4

C 1 and 4 only

D 2 and 3 only

19 In most organisms, six different triplets of the DNA strand that is complementary to mRNA code for the amino acid serine: AGA, AGG, AGT, AGC, TCA and TCG.

In the yeast *Candida albicans*, a seventh DNA triplet, GAC, also codes for serine. In most organisms, this triplet codes for leucine.

The diagram shows part of an mRNA molecule from *C. albicans*.

	– AGU –	UCG –	CGG –	UCA –	AGC –	ACC –	UGG –
codon number	– 11 –	12 –	13 –	14 –	15 –	16 –	17 –

Which mutation of the DNA that is complementary to this mRNA could result in *C. albicans* producing a polypeptide with an unbroken sequence of five serines in it?

A substituting a purine with a pyrimidine in the DNA coding for codon 13

B substitutinq a purine with a pyrimidine in the DNA coding for codon 16

C substituting a pyrimidine with a purine in the DNA coding for codon 13

D substituting a pyrimidine with a purine in the DNA coding for codon 16

20 The diagram shows the banding pattern of two human chromosomes. **P** is a normal chromosome.

What accounts for chromosome **Q**?

A crossing over between sister chromatids

B deletion of part of the chromosome

C inversion of part of the chromosome

D translocation of part of another chromosome

21 Albino horses have recessive alleles which cannot be transcribed to produce pigmentation. Geneticists studying horses have found a different allele of the same gene which predisposes carriers to premature aging of their follicles. This allele is dominant, so, even when born with pigmented hair, the coats of horses with this allele will gradually turn white over the first 6 to 8 years of life.

Which **white** female horse would be the best to use in crosses to determine the genotype of a ten-year-old, white male horse?

A an unrelated, ten-year-old female

B an unrelated, three-year-old female

C a sister of the male

D the male's mother

22 *Hydrangea macrophylia* bushes can produce flowers of different colours: white, pink or blue. Bushes with white flowers always produce white flowers. However, the flowers of bushes with pink or blue flowers can change colour. As soil pH increases, flower colour changes from pink to blue and this is reversed as soil pH falls.

Which statement describes the control of flower colour in *H. macrophylia*?

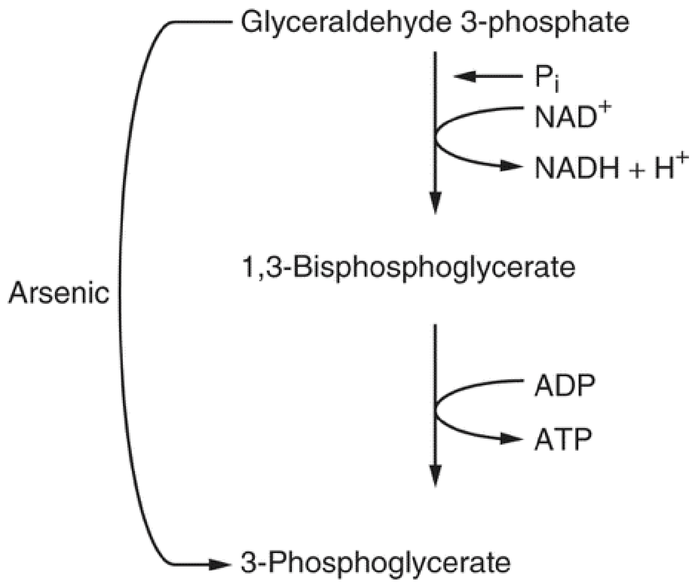
A The flower colour of bushes with white flowers depends on genes and the environment. The flower colour of bushes with pink or blue flowers depends only on the environment.

B The flower colour of bushes with white flowers depends only on genes. The flower colour of bushes with pink or blue flowers depends on genes and the environment.

C The flower colour of bushes with white flowers depends on genes and the environment. The flower colour of bushes with pink or blue flowers depends on genes and the environment.

D The flower colour of bushes with white flowers depends only on genes. The flower colour of bushes with pink or blue flowers depends only on the environment.

23 The diagram shows the effect of arsenic on the metabolism of glyceraldehyde-3-phosphate.



What is the net yield of ATP molecules from the glycolysis process involving 2 molecules of glucose in the presence of arsenic?

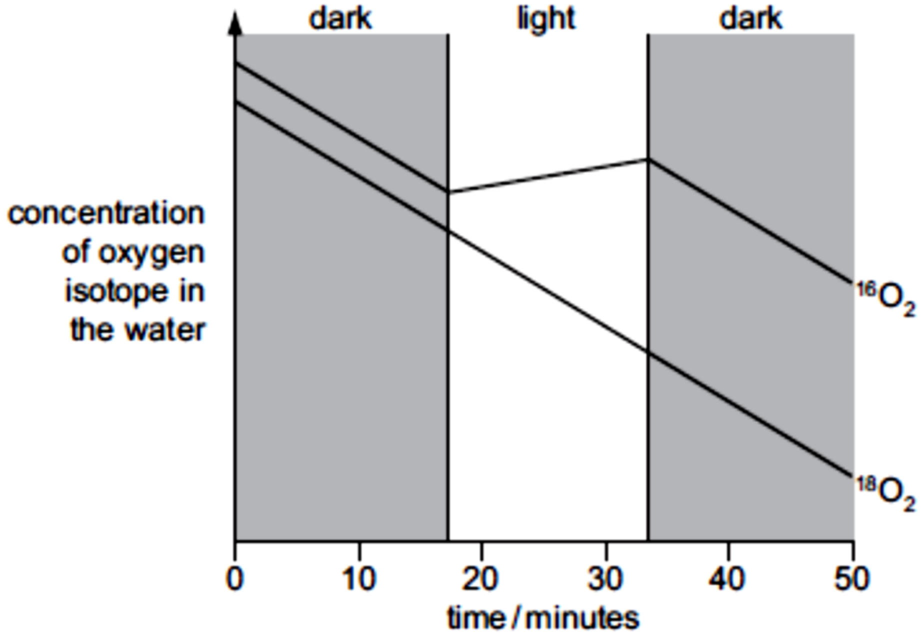
A 0

B 1

C 2

D 4

24 The common isotope of oxygen is ^{16}O. Air containing $^{16}O_2$ and $^{18}O_2$ was bubbled through a suspension of algae for a limited period. After this, the concentration of these two isotopes of oxygen in the water was monitored for the next 50 minutes whilst the algae were subjected to periods of dark and light. The results are shown in the diagram.



What is the best explanation for these results?

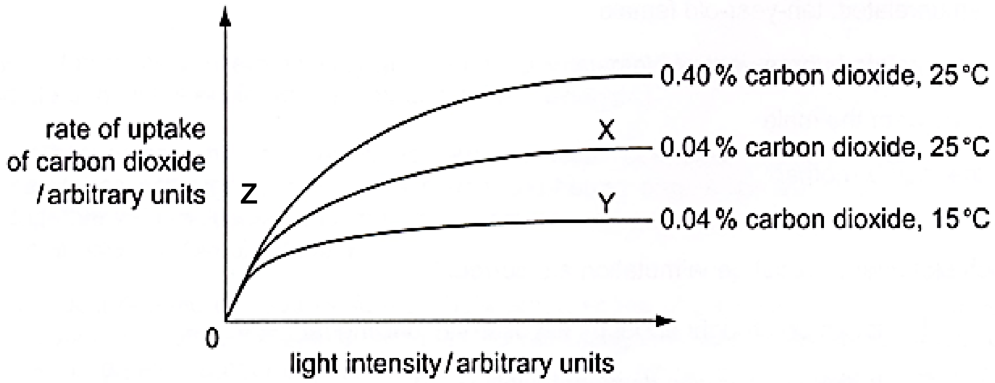
A Both isotopes of oxygen are used by the algae in the dark in respiration, but in the light oxygen is produced from water in photorespiration.

B The algae can distinguish chemically between the two isotopes.

C The algae produce oxygen from the water used in photosynthesis, but only in the light.

D The two isotopes have different rates of diffusion.

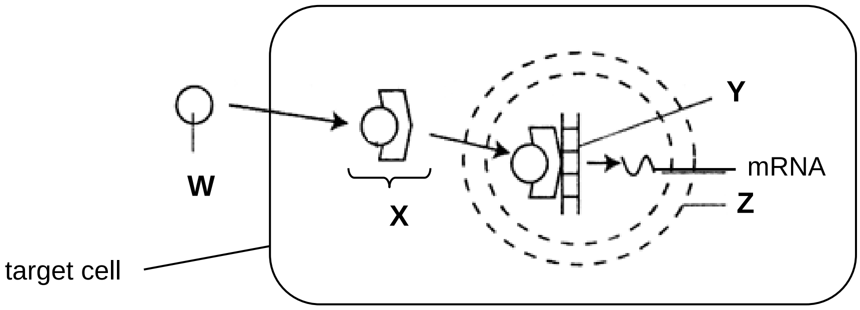
25 The graph shows the rate of uptake of carbon dioxide by a photosynthetic plant in different conditions.



What are the limiting factors at points **X**, **Y** and **Z**?

	X	Y	Z
A	carbon dioxide concentration	temperature	light intensity
B	temperature	carbon dioxide concentration	light intensity
C	carbon dioxide concentration	light intensity	temperature
D	light intensity	temperature	carbon dioxide concentration

26 The diagram below shows the action of a hormone.



Which of the following correctly identifies **W** to **Z**?

	W	X	Y	Z
A	insulin	receptor tyrosine kinase	insulin	secretory vesicle
B	glucagon	G-protein coupled receptor	cAMP	protein kinase
C	oestrogen	second messenger	oestrogen receptor	cell surface membrane
D	testosterone	testosterone-receptor complex	DNA	nuclear membrane

27 Bacteria in the genus *Wolbachia* infect many species of insects. The bacteria live as parasites within the cells of their insect hosts and are passed from one generation of host to the next through the eggs of infected females. Infected males cannot pass the infection on to the next generation, since sperm are too small to be parasitised by *Wolbachia*.

Wolbachia has evolved several strategies to increase the probability that the infection will be passed on from one generation of host to the next. These include:

- feminisation, in which infected males become fertile females
- male killing, in which male embryos that are developing from infected fertilised eggs die
- parthenogenesis, in which infected females become capable of reproducing asexually
- cytoplasmic incompatibility, in which infected males are unable to reproduce successfully with uninfected females.

Which of these statements could explain how these strategies benefit *Wolbachia* thorugh natural selection?

1 Feminisation increases the number of insects infected with *Wolbachia* that can pass on the infection to their offspring.

2 Male killing decreases the proportion of uninfected individuals in the offspring of an infected female.

3 Parthenogenesis increases the probability that an infected female will pass on the infection to its offspring, since there is no need to find a mate.

4 Cytoplasmic incompatibility increases the fitness of infected females by allowing infected females to reproduce with uninfected males.

A 1, 2 and 3

B 1, 2 and 4

C 1 and 3 only

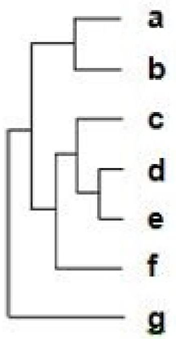
D 2 and 4 only

28 The table shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

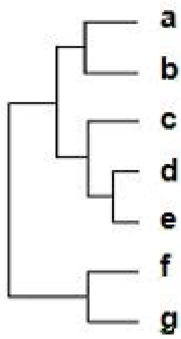
	b	**c**	**d**	**e**	**f**	**g**
a	39	72	128	126	127	269
b		81	130	128	129	268
c			129	127	128	267
d				56	154	271
e					151	268
f						273

Which of the following phylogenetic trees best shows the relationship among these seven species?

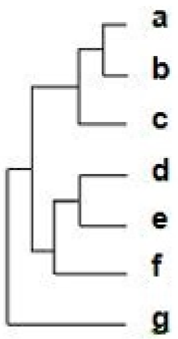
A



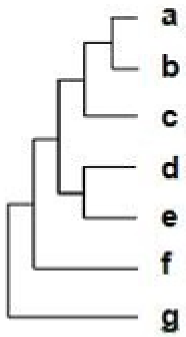
B



C

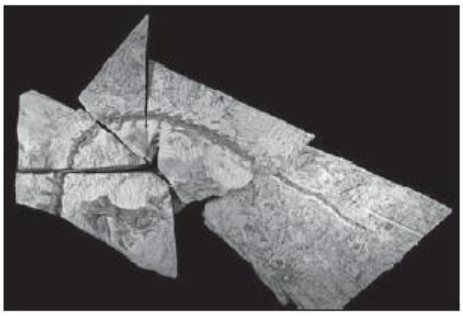


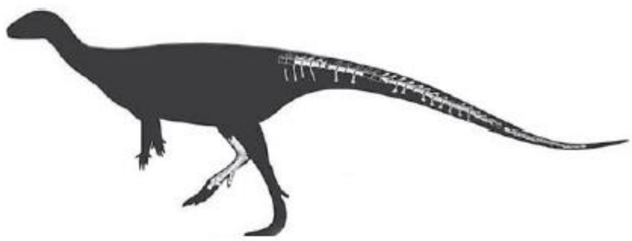
D



29 The photograph on the left below shows a fossil of the ornithopod dinosaur *Diluvicursor pickeringi* found in 113-million-year-old rock in western Victoria, Australia. The fossil consists of a tail, a partial hind limb and some vertebrae. *D. pickeringi* grew to 2.3 m long.

The diagram on the right below is a reconstruction of the complete dinosaur. Evidence suggests that the dinosaur was fossilised in a log-filled hollow at the bottom of an ancient riverbed. Two younger fossils that had been found previously at a nearby site are closely related to *D. pickeringi*.





Palaeontologists believe that the Victorian ornithopods shared a close common ancestor with several ornithopod fossils found in Antarctica, South America and Africa.

Which one of the following is the most likely explanation for the distribution of these fossils?

A Antarctica, South America and Africa were joined to Australia in the distant past.

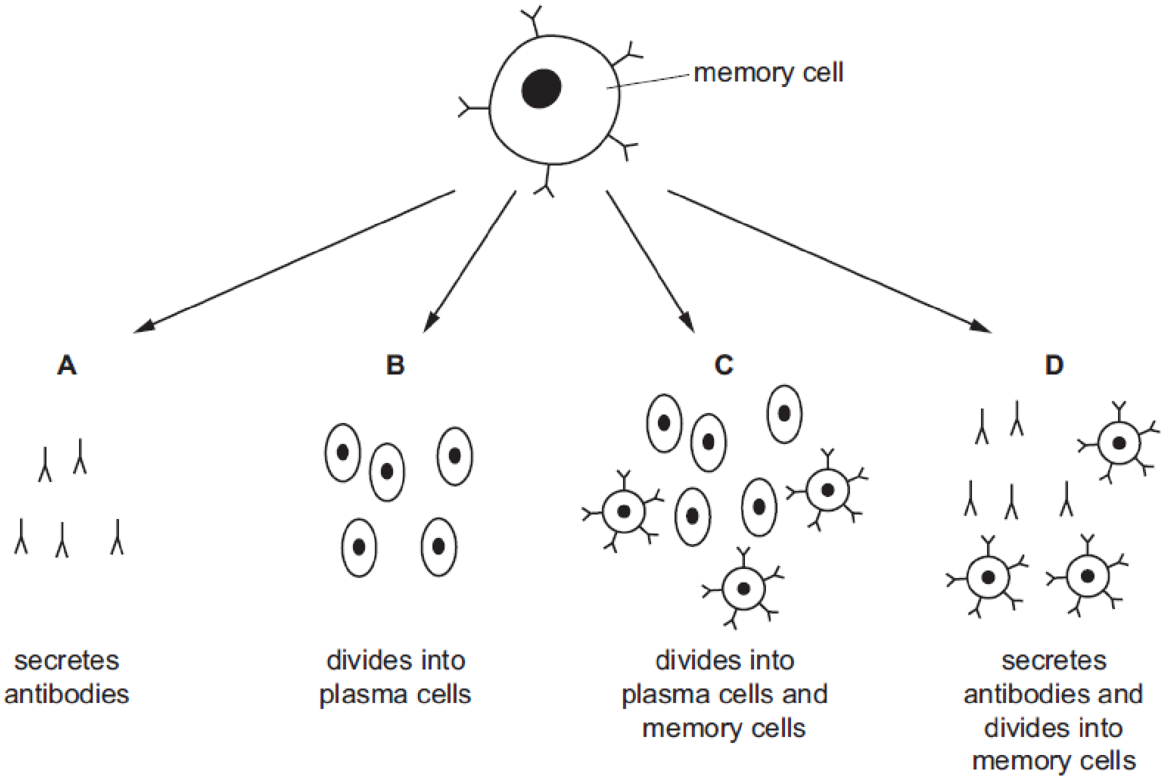
B Seagoing, scavenger birds carried the fossil bones of the ornithopods to other continents.

C The small forelimbs of the ornithopods suggest that they were evolving wings for flight.

D The strong tails of the ornithopods enabled them to swim for sustained periods of time.

30 When exposed to an antigen for a second time, memory cells stimulate a secondary immune response.

Which correctly shows the secondary immune response?



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